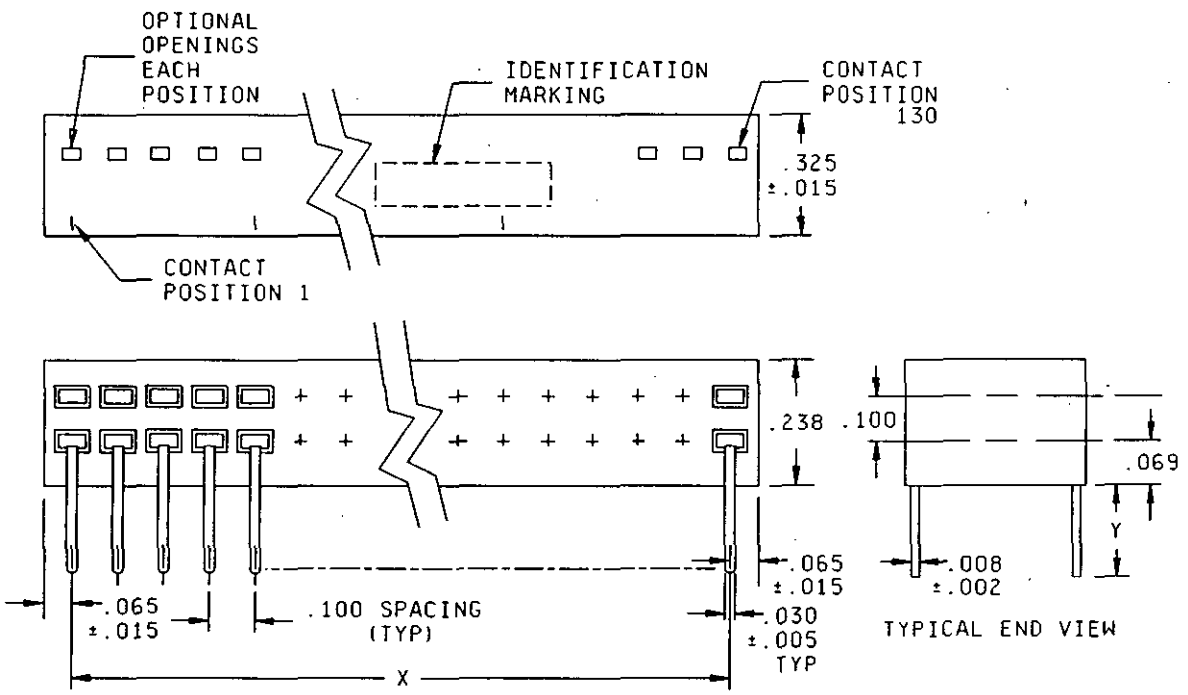
INCH-POUND
MIL-C-55302/128C(USAF)
6 MAY 1993
SUPERSEDING
MIL-C-55302/128B(USAF)
21 September 1988

MILITARY SPECIFICATION SHEET

# CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES: RECEPTACLE, DOUBLE ROW, 4 THROUGH 130 CONTACT POSITIONS, FOR PRINTED WIRING BOARDS (1/16", 3/32", AND 1/8") (.100 SPACING)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-55302.



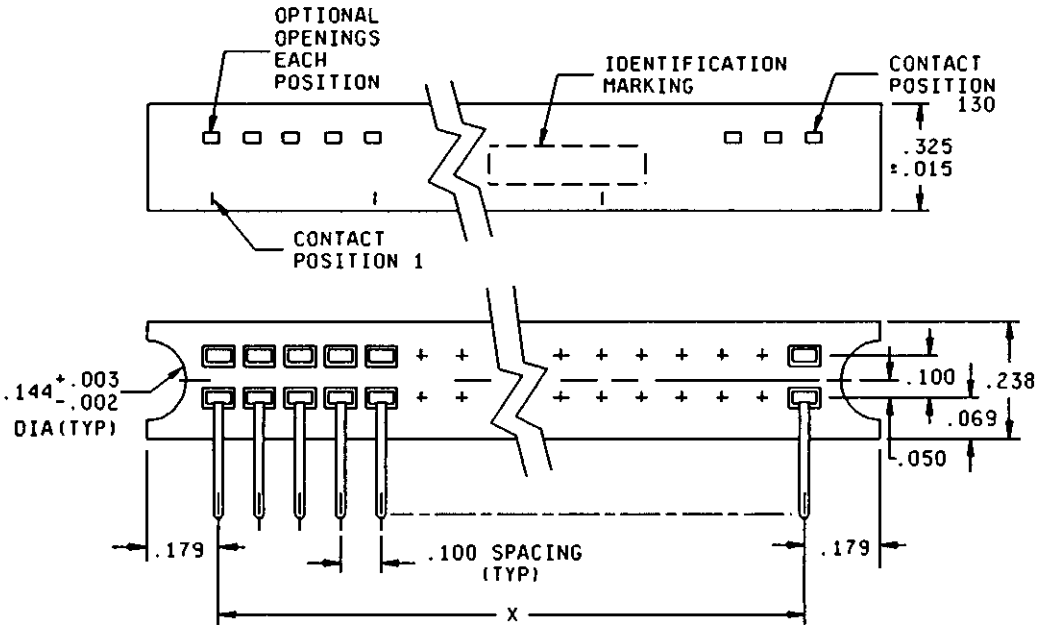

HOUSING CONFIGURATION 1

Ⓒ FIGURE 1. Connector, receptacle, .100 (2.54 mm) spacing.

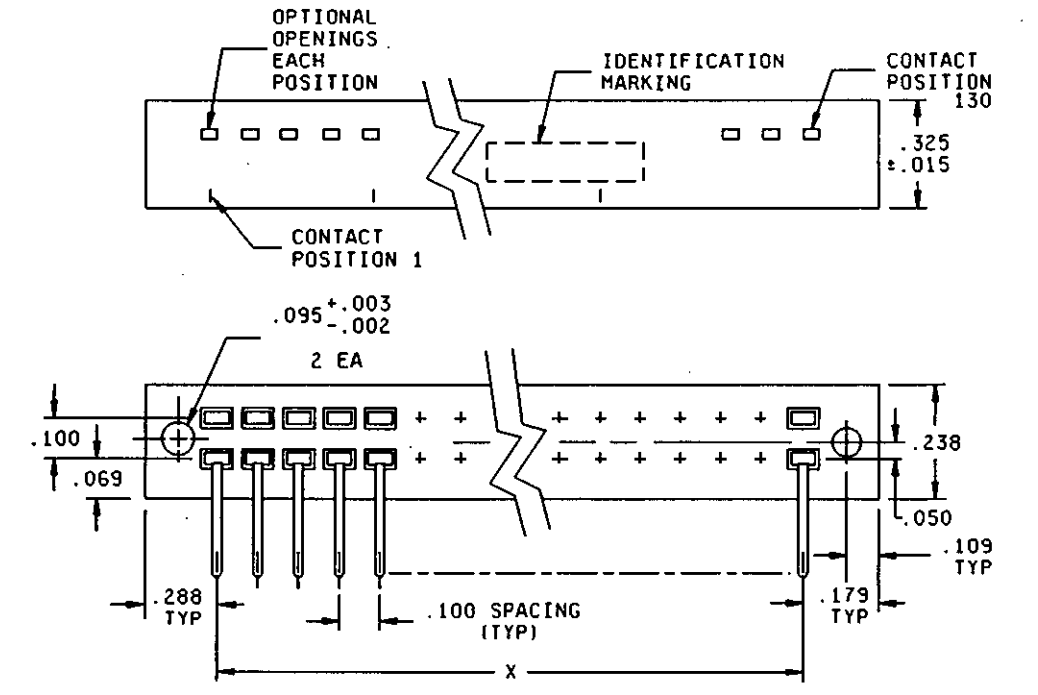
Ⓒ denotes changes.


DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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HOUSING CONFIGURATION 2

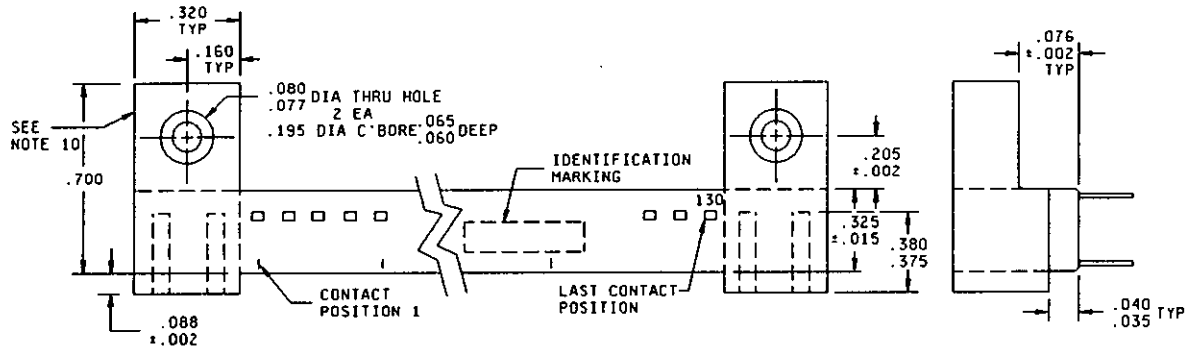


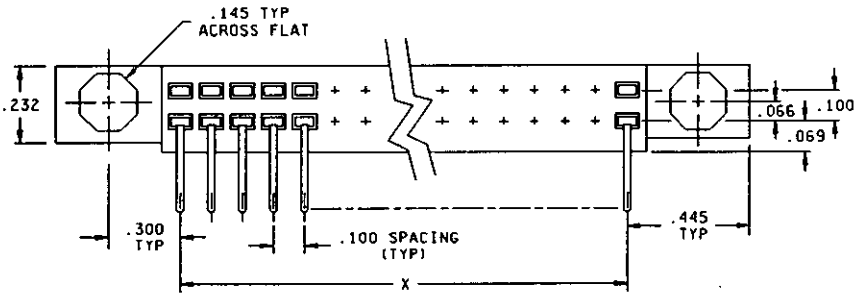

HOUSING CONFIGURATION 3

© FIGURE 1. Connector, receptacle, .100 (2.54 mm) spacing - Continued.

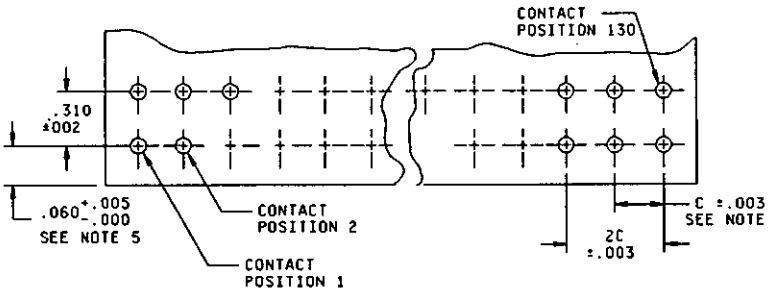


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HOUSING CONFIGURATION 4




RECOMMENDED PRINTED
CIRCUIT BOARD HOLE LAYOUT

NOTE: Dimension C, contact centerline, to be .100 (2.54 mm); ±.003 (0.08) tolerances not to accumulate within one connector pattern.

© FIGURE 1. Connector, receptacle, .100 (2.54 mm) spacing.



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| Inches | mm | Inches | mm | Inches | mm | Inches | mm |
|---|---|---|---|---|---|---|---|
| .002 | 0.05 | .066 | 1.68 | .145 | 3.68 | .300 | 7.62 |
| .003 | 0.08 | .069 | 1.75 | .160 | 4.06 | .310 | 7.87 |
| .005 | 0.13 | .076 | 1.93 | .179 | 4.55 | .320 | 8.13 |
| .015 | 0.38 | .077 | 1.96 | .195 | 4.95 | .325 | 8.26 |
| .035 | 0.89 | .080 | 2.03 | .206 | 5.21 | .375 | 9.52 |
| .040 | 1.02 | .088 | 2.24 | .232 | 5.89 | .380 | 9.65 |
| .050 | 1.27 | .095 | 2.41 | .238 | 6.05 | .445 | 11.30 |
| .060 | 1.52 | .100 | 2.54 | .288 | 7.32 | .700 | 17.78 |
| .065 | 1.65 | .144 | 3.66 | | | | |

NOTES

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. The mating post array must be maintained on .100 centerline grid spacing.
5. Post engagement length is .323 maximum, .308 minimum from edge of board, when connector is mounted in accordance with figure 1. Total mating positions to be determined by connector size in accordance with table I.
6. Post location tolerance not to accumulate.
7. Mating post tips must be maintained within .015 diameter true position.
8. Mating posts may consist of posts staked directly into printed circuit boards, or mounted in insulator bodies.
9. The first and last circuit shall be numbered and every fifth circuit shall have an identifiable mark on the top surface of the housing beginning with the first circuit on the left (no. 1) looking at the mating face of the connector.
10. Mounting ears shall be molded or bonded to the insulator.
11. Housing configurations 1, 3, and 4 terminate directly to the board and therefore do not meet the requirements of MIL-STD-275.
12. Housing configuration 2 mates with M55302/165 and does meet the requirements of MIL-STD-275 for two piece connectors.

(C) FIGURE 1. Connectors, receptacle, .100 (2.54 mm) spacing - Continued.



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TABLE I. Dimensions. 1/

| Code | Number of contacts | Dimension X | Code | Number of contacts | Dimension X | Code | Number of contacts | Dimension X | Code | Number of contacts | Dimension X |
|---|---|---|---|---|---|---|---|---|---|---|---|
| AA | 4 | .100 (2.54) | BA | 36 | 1.700 (43.18) | CA | 68 | 3.300 (83.82) | DA | 100 | 4.900 (124.46) |
| AB | 6 | .200 (5.08) | BB | 38 | 1.800 (45.72) | CB | 70 | 3.400 (86.36) | DB | 102 | 5.000 (127.00) |
| AC | 8 | .300 (7.62) | BC | 40 | 1.900 (48.26) | CC | 72 | 3.500 (88.90) | DC | 104 | 5.100 (129.54) |
| AD | 10 | .400 (10.16) | BD | 42 | 2.000 (50.80) | CD | 74 | 3.600 (91.44) | DD | 106 | 5.100 (132.08) |
| AE | 12 | .500 (12.70) | BE | 44 | 2.100 (53.34) | CE | 76 | 3.700 (93.98) | DE | 108 | 5.300 (134.62) |
| AF | 14 | .600 (15.24) | BF | 46 | 2.200 (55.88) | CF | 78 | 3.800 (96.52) | DF | 110 | 5.400 (137.16) |
| AG | 16 | .700 (17.78) | BG | 48 | 2.300 (58.42) | CG | 80 | 3.900 (99.06) | DG | 112 | 5.500 (139.70) |
| AH | 18 | .800 (20.32) | BH | 50 | 2.400 (60.96) | CH | 82 | 4.000 (101.60) | DH | 114 | 5.600 (142.24) |
| AJ | 20 | .900 (22.86) | BJ | 52 | 2.500 (63.50) | CJ | 84 | 4.100 (104.14) | DJ | 116 | 5.700 (144.78) |
| AK | 22 | 1.000 (25.40) | BK | 54 | 2.600 (66.04) | CK | 86 | 4.200 (106.68) | DK | 118 | 5.800 (147.32) |
| AL | 24 | 1.100 (27.94) | BL | 56 | 2.700 (68.58) | CL | 88 | 4.300 (109.22) | DL | 120 | 5.900 (149.86) |
| AM | 26 | 1.200 (30.48) | BM | 58 | 2.800 (71.12) | CM | 90 | 4.400 (111.76) | DM | 122 | 6.000 (152.40) |
| AN | 28 | 1.300 (33.02) | BN | 60 | 2.900 (73.66) | CN | 92 | 4.500 (114.30) | DN | 124 | 6.100 (154.94) |
| AP | 30 | 1.400 (35.56) | BP | 62 | 3.000 (76.20) | CP | 94 | 4.600 (116.84) | DP | 126 | 6.200 (157.48) |
| AQ | 32 | 1.500 (38.10) | BQ | 64 | 3.100 (78.74) | CQ | 96 | 4.700 (119.38) | DQ | 128 | 6.300 (160.02) |
| AR | 34 | 1.600 (40.64) | BR | 66 | 3.200 (81.28) | CR | 98 | 4.800 (121.92) | DR | 130 | 6.400 (162.56) |

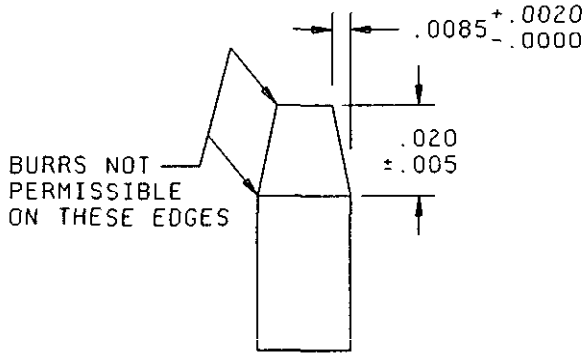
1/ Metric equivalents are in parentheses.



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(C) TABLE II. Printed circuit board and plating thickness.

| Code | Board thickness | Y dimension | Contact area plating thickness | *Remainder* plating thickness |
|---|---|---|---|---|
| A | .062 | .115 | Class 1 | Flash |
| B | .062 | .115 | Class 1 | Class 1 |
| C | .093 | .145 | Class 1 | Flash |
| D | .093 | .145 | Class 1 | Class 1 |
| E | .125 | .175 | Class 1 | Flash |
| F | .125 | .175 | Class 1 | Class 1 |
| G | .062 | .115 | Class 1 | Tin/lead dipped or plated |
| H | .093 | .145 | Class 1 | Tin/lead dipped or plated |
| I | .125 | .175 | Class 1 | Tin/lead dipped or plated |




| Inches | mm |
|---|---|
| .0020 | 0.051 |
| .005 | 0.13 |
| .0085 | 0.216 |
| .020 | 0.51 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 2. Mating post tip configuration.



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REQUIREMENTS:

Design and construction:

Dimensions and configurations: See figures 1 and 2 and tables I and II.

Material:

Body: In accordance with MIL-C-55302.

Plating: The contact plating shall be gold in accordance with MIL-C-55302 as specified in table II, over nickel plating in accordance with MIL-C-55302.

Contact cavities: See figure 1.

Printed wiring terminations: See figure 1.

Contact identification: See figure 1.

Polarization: Connector polarization may be accomplished by use of a keying plug (P/N M55302/127-00KY) which is inserted into the particular contact circuit to be keyed. Keying plug is nylon in accordance with ASTM D-4066.

Color: Natural.

Alignment: To be accomplished by user.

Method of mounting: Not applicable.

Oversized pin exclusion (sockets only): Shall not accept a .050 inch diameter pin.

Contact engagement and separation forces:

Engagement force: Shall be 6 ounces maximum after three insertions of a .0260 +.0000, -.0001 inch square pin.

Separation force: Shall be .75 ounce minimum.

Mating and unmating: The maximum connector mating force shall be 7 ounces times the number of contact positions, and the minimum connector unmating force shall be .75 ounce times the number of contact positions. For purposes of testing, the mating plug connector shall be considered to be the correct number of .025 wrappost in accordance with MIL-STD-1130, gold-plated. Maximum post engagement length is .323 inch and minimum post engagement length is .308 inch, when connector is mounted as shown on figure 1.

Mating post: See figure 2. For total mating positions, see table I.

Contact rating: 3.0 amperes maximum.

Contact resistance: No individual contact pair shall have a resistance exceeding 12 milliohms when mated to a gold-plated .025 wrappost in accordance with MIL-STD-1130.

Contact retention: Not applicable.

Dielectric withstanding voltage:

Sea level: 750 volts rms.

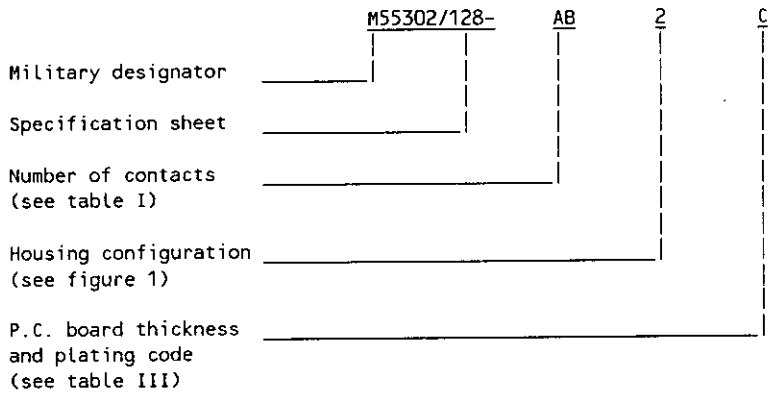
High altitude: 275 volts rms.



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Contact bend test: The body of the contact shall be slowly bent through 90° and then returned to the original position. A bend through 90° and return to original position shall be defined as one bend. The test shall include three bending cycles.

Part number: Shall be as shown in the following example:




## CONCLUDING MATERIAL

Custodian:
Army - CR
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - AS, OS, SH
Air Force - 11, 85
DLA - ES

User activities:
Army - AT, AV, ME
Navy - MC
Air Force - 19

Preparing activity:
Air Force - 17

Agent:
DLA - ES

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